

TPG4190 Seismic data acquisition and processing

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Information

1. Learning objectives
2. Schedule
3. Material
4. Exercises
5. Project
6. Exam
7. Content

Learning objectives

- ▶ **Know** basic theory and principles of seismic data acquisition
- ▶ **Know** key steps in seismic processing
- ▶ **Know** how to derive and apply basic equations that are used in acquisition and processing of seismic data.
- ▶ **Know** how to perform simple seismic processing of seismic field data.

Schedule

- ▶ **Lecture** Every Tuesday 12:15-14:00, PTS1 room P12
- ▶ **Lecture** Every Wednesday 08:15-10:00, PTS1 room P13
- ▶ **Exercise** Every Thursday 08:15-09:00, PTS1 room P12

Material

- ▶ Lecture slides 1-16
- ▶ Lecture notes (Martin Landrø)

Material found at: **`git@github.com:barntsen/Seislectures`**

Exercises

- ▶ No obligatory exercises
- ▶ Exercises (and solutions) provided

Project

- ▶ Obligatory project
- ▶ Processing of Regional Seismic data set from the Pacific
- ▶ Sherwater Reveal processing software
- ▶ Takes place in in the computerlab room 4, PTS1 every Thursday 08:15-10:00. Start September 4 (date may change).
- ▶ Deliver report by 15. November

Exam

- ▶ School exam
- ▶ No aids allowed
- ▶ Grade A-F
- ▶ Project have to be approved

Content

1. Acquisition of seismic data
2. Modeling of seismic data
3. Preprocessing and Noise
4. Imaging of seismic data
5. Multiple removal
6. Tomography
7. Full Waveform Inversion
8. Exam

Acquisition of seismic data

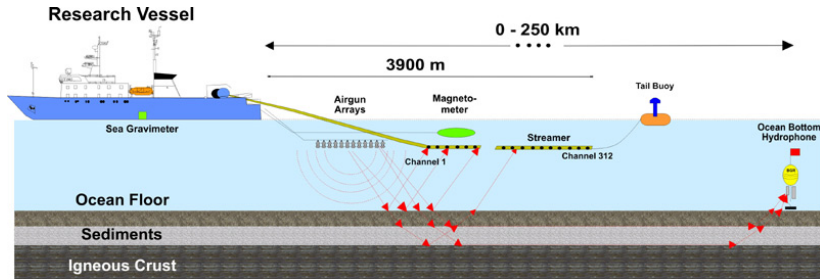


Figure: Seismic ship

Modeling of seismic data

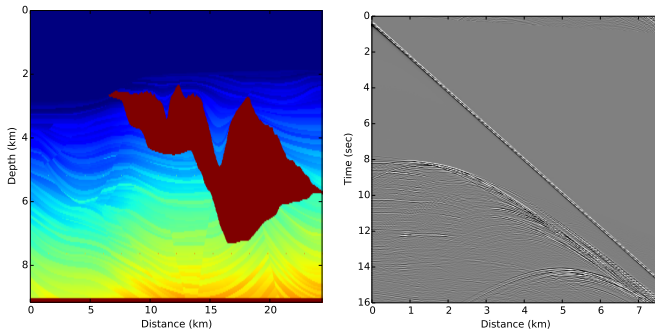
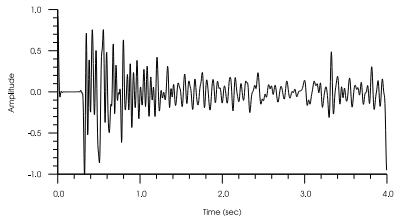
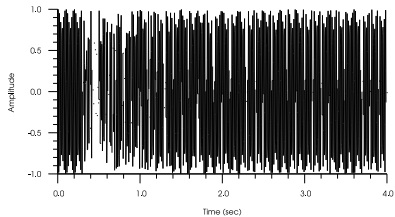


Figure: Input model (left) and Output data (right)

Movie file: snp.mp4

Preprocessing and Noise



Imaging of Seismic data

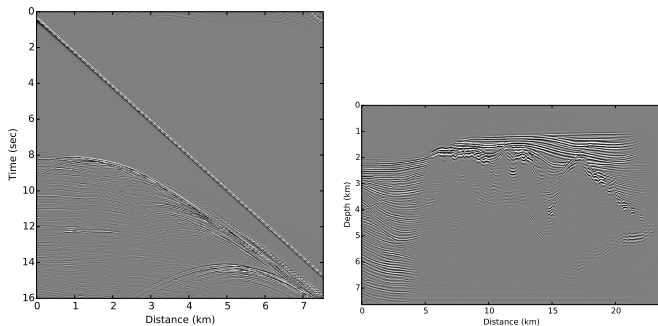
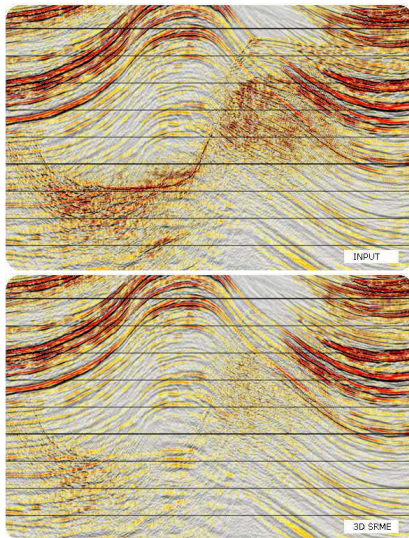
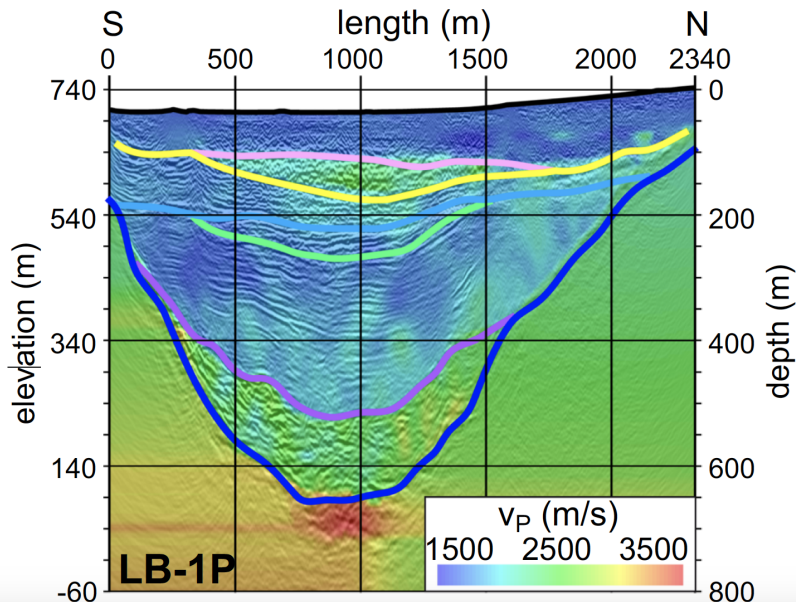


Figure: Input data (left) and Output migration (right)

Multiples



Tomography



Inversion

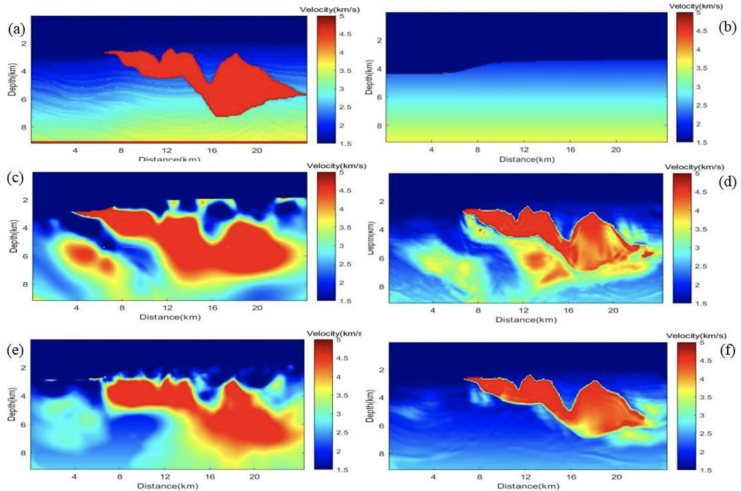


Figure 3. (a) Sigsbee2A model; (b) Initial model; (c) Inversion result of MSDEI; (d) Inversion result of MSDEI+FWI; (e) Inversion result of S-MSDEI; (f) Inversion result of S-MSDEI+FWI.

└ Inversion

Inversion

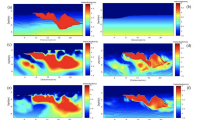


Figure 3. (a) Sigsbee24 model; (b) Inversion result of M0D02; (c) Inversion result of M0D02 + FWI; (d) Inversion result of S-00D02; (e) Inversion result of S-00D02 + FWI; (f) Inversion result of S-00D02 + FWI.

Full waveform inversion of the Sigsbee test model